Smart Haptic Glove with sensory displacement and dexterity measurement

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Introduction

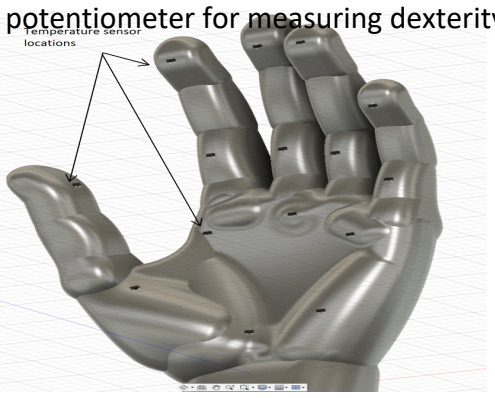
Loss or diminished capacity of a sensory organ has always been a concern as they seriously effect every day chores.An example of lost sensation is the peripheral nerve damage in the disease Leprosy caused by Mycobacterium leprae. In hands the loss in sensation makes the patient lose the perception of Pain, touch and temperature. The loss in sensation of temperature leads to burns and the surface of the hands are damaged.Gloves have been made for the perception of touch/pressure to reduce damage but due to the glove and the already severally dexterously challenged hands its not practical for everyday chores. Additionally continuous improvement data for dexterity is not available.

SCOPE of the Project

The main objective is to present a four-legged robot (quadruped) that can walk with stability on plane and uneven surfaces with focus on minimizing the torque borne by the motors. The control system for the motor used is designed and adjusted to give smooth response with minimum steady state error. The forward and inverse kinematics have been laid followed by trajectory planning. Gait study has been done and stability criteria have been explored. Finally torque optimization has been done with particular focus on the variables: step length and descent of torso of the robot.

Methodology

The Glove is designed for ease in wearability and low manufacturing cost. Each palm contains 15 temperature sensors as shown in figure 1 to detect temperature of objects that are touched by the user and relay the information to a Microcontroller for processing. The back of the hand provides clamps and guides for wires that will be connected to the potentiometer for measuring dexterity.



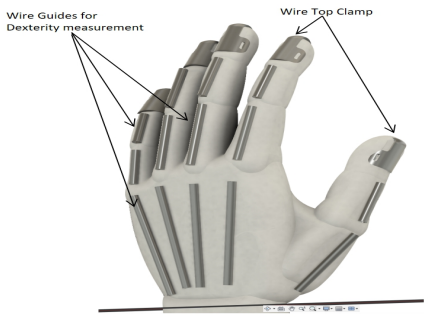


Figure 1

The data from the sensors and potentiometers are received and processed by the microcontroller and sentover WiFi to the data base for futher processing

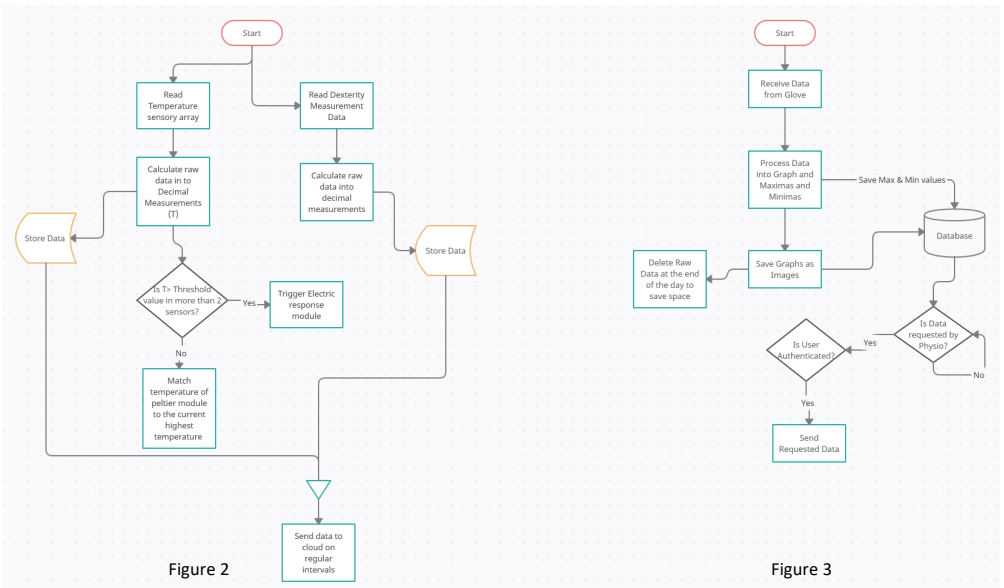


Figure 2

Figure 3

The base of the arm contains the Response modules triggered by events detected by the Microcontroller. The Electric trigger(Figure 4) module provides with a quick response to the user to drop what ever the user is holding. The thermal feedback(Figure 5) provides the sensory displacement to feel warm objects as they would appear on the fingers.

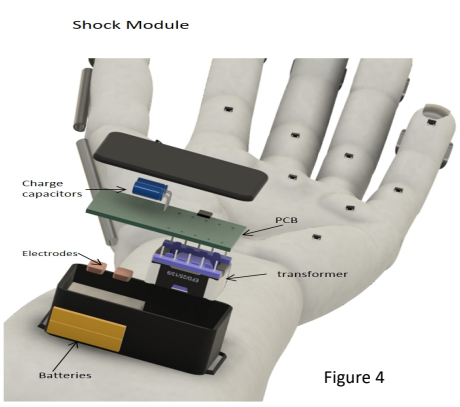


Figure 4

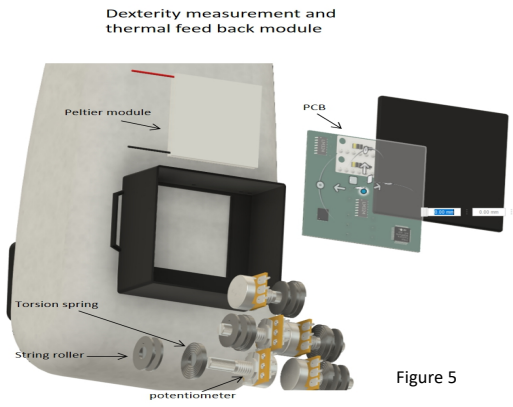


Figure 5

Results

yet to be added

yet to be added

Conclusion

yet to be added

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